Soo Min Lee

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EDUCATION

University of California San Diego (UCSD) CA, U.S.A.
Ph.D., Major in Bioinformatics and Systems Biology *September 2025 – Present*
Korea Advanced Institute of Science and Technology (KAIST) Daejeon, Korea
M.S., Major in Bio and Brain Engineering *August 2020 – August 2022*
Handong Global University (HGU) Pohang, Korea
B.S., Double major in Computer Science and Life Science *Feb 2017 – August 2020*
Summa cum laude, Early graduation

RESEARCH & PROFESSIONAL EXPERIENCE

La Jolla Institute for Immunology CA, U.S.A.
Graduate student researcher, Laboratory of Tal Einav September 2025 – Present
Inocras (Precision oncology start-up) Daejeon, Korea
Computational Genomics Center – Pipeline Development Team March 2023 – June 2025

- *Key problem*: How can we rapidly translate complex cancer genomics into precise, actionable treatments?
- Developed algorithms for targetable mutation calling, drug recommendations, and trial matching by integrating Cancer Knowledgebase (CKB) data into somatic pipelines.
- Refined structural variation (SV) algorithms, reducing false-positive calls by up to 78% while accelerating pipeline speed.
- Built an AI-driven deep mutation caller for small variants, enhancing the accuracy and efficiency of mutation detection.

Korea Advanced Institute of Science and Technology Daejeon, Korea
Graduate student researcher, Laboratory of Kwanghyun Cho July 2020 – Feb 2023

- *Key problem*: How do lung cancer cells develop resistance to p53 reactivation treatments?
- Developed a novel computational approach using Generative adversarial networks and Graph neural networks to identify drug resistance mechanisms and therapeutic targets.
- Identified a key pathway and biomarker associated with p53 resistance and validated the therapeutic potential of the identified gene targets via *in vitro* experiments.

RESEARCH HIGHLIGHTS

Understanding Early-Life Immunity. How do transferred maternal antibodies decay and shape an infant's response to subsequent viral exposures and vaccinations? By applying dynamics modeling to a longitudinal cohort tracked across an infant's first two years of life, this work provides a high-resolution map of how maternal immune status dictates early-life antibody kinetics and durability, laying the computational foundation to optimize both maternal and infant vaccination timing.

Optimizing Early-Life Diagnostics. How can we detect viral infections in early life with maximum accuracy while minimizing assay complexity and cost? We systematically evaluate multi-assay diagnostic frameworks, comparing hemagglutination inhibition (HAI) titers, multiplexed MSD IgG/IgA, anti-nucleoprotein (NP), and RT-PCR from nasal swabs, to define the most accurate and cost-effective surveillance strategies for infant infections.

Mapping Proteome-Wide Immunity. How do protective immune responses emerge in infancy across multiple viral exposures? Utilizing high-throughput phage display immunoprecipitation sequencing (PhIP-seq), this work characterizes peptide-level antibody specificity changes following viral vaccination and infection to uncover the key features dictating immune trajectories and durability.

PUBLICATIONS & PATENTS

1. **Lee SM**, Burrell AR, Spranger S, et al. The Birth of Influenza Immunity: High-Resolution Antibody Dynamics Driven by Maternal Antibody Waning, Vaccinations, and Infections during the first Two Years of Life. *medRxiv* [Preprint]. 2026 May 6. doi: 10.64898/2026.05.04.26352413. (Under review at *Nature Immunology*).
2. **Lee SM**, Han Y, Cho KH. Deep learning untangles the resistance mechanism of p53 reactivator in lung cancer cells. *Cell iScience*. 2023 Nov 1;26(12):108377. doi: 10.1016/j.isci.2023.108377. PMID: 38034356; PMCID: PMC10682260.
3. USES OF THBS1 INHIBITOR FOR OVERCOMING DRUG RESISTANCE OF CANCER. Application: 10-2023-0038078, 2023 March 23 (Republic of Korea), PCT/KR2023/003905, 2023 March 23 (PCT), 18/840,605, 2024 August 22 (USA)

AWARDS

- SPARK Award Finalist, La Jolla Institute for Immunology 2026
- Top honor in Life Science (1%), Handong Global University *2020*
- Academic Excellence Scholarship, Handong Global University *2018, 2019, 2020*